



OPEN Mathematical analysis of an anaerobic digestion model for biogas production from solid waste

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Municipal solid waste is one of the most significant sources of gas emissions. In a context of renewable energy production and greenhouse gas emission reduction, we focus on the amount of biogas produced by the anaerobic digestion of organic matter in a controlled landfill environment. We present a mathematical model describing this process from a microbiological and biochemical point of view, including the dynamics of the methane and carbon dioxide produced. A qualitative analysis of the dynamic system shows the existence of an infinite number of non-hyperbolic equilibria inducing an attractor that has been identified. Through numerical tests, we explore how the non-connectivity of the attractor resulting from the choice of growth functions can entirely transform the performance of the process, and highlight critical initial stock values that influence the amount of biogas produced.

Keywords Biogas production, Anaerobic digestion, Dynamical systems theory, Stable manifold theorem, Optimal initial stock

List of symbols

S_s	Total concentration of biodegradable organic substrate (g/l)
S_{dc}	Dissolved organic substrate concentration (g/l)
X_a	Acidogenic biomass concentration (g/l)
S_{ac}	Acetate substrate concentration (g/l)
X_m	Methanogenic biomass concentration (g/l)
CH_4	Methane concentration (g/l)
CO_2	Carbon dioxide concentration (g/l)
Y	Vector of variables
y_a	Mass conversion rate of dissolved organic substrate into acidogenic biomass
y_m	Mass conversion rate of acetate substrate into methanogenic biomass
y_{ac}	Acetate substrate conversion yield y_{CH_4} Methane conversion yield
K_d	Half-saturation constant of dissolved organic substrate (g/l)
K_a	Half-saturation constant of acetate substrate (g/l)
K_{id}	Inhibition constant of dissolved organic substrate (g/l)
K_{ia}	Inhibition constant of acetate substrate (g/l)
k_h	Hydrolysis constant of biodegradable organic substrate (day^{-1})
d_a	Death rate of acidogenic bacteria (day^{-1})
d_m	Death rate of methanogenic bacteria (day^{-1})
μ_a^{max}	Maximum growth rate of acidogenic bacteria (day^{-1})
μ_m^{max}	Maximum growth rate of methanogenic bacteria (day^{-1})
t	Time (day)
S_{dc}^*	Limit concentration of dissolved organic substrate (g/l)
S_{ac}^*	Limit concentration of acetate substrate (g/l)
$\overline{CH_4}$	Limit concentration of methane (g/l)
$\overline{CO_2}$	Limit concentration of carbon dioxide (g/l)
$\mu_a(\cdot)$	Growth function of acidogenic bacteria
$\mu_m(\cdot)$	Growth function of methanogenic bacteria

The bio-degradation process of municipal waste in landfills consists of a succession of chemical reactions catalyzed by bacteria, which transform slowly the organic matter contained in the waste into mineral or gaseous

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elements. This transformation of organic matter can be decomposed in many steps, which differ depending on the environment being under aerobic or anaerobic conditions. The aerobic digestion lasts only a few days after the waste is deposited in the landfill, where oxygen is present in the void spaces. During this period, the biogas produced is mainly carbon dioxide. The anaerobic digestion operates within a closed environment where, in the absence of oxygen, organic matter is gradually converted into biogas, primarily composed of carbon dioxide and methane^{1,2}. This process unfolds in four major stages: first, hydrolysis breaks down insoluble organic polymers into soluble derivatives that become accessible to other bacteria. Next, during acidogenesis, acidogenic bacteria convert these sugars and amino acids into carbon dioxide, hydrogen, ammonia, and organic acids. In the acetogenesis stage, these organic acids are further processed by bacteria into acetic acid, along with additional ammonia, hydrogen, and carbon dioxide. Finally, methanogens convert these end products into methane and carbon dioxide, completing the biogas production process^{3,4}.

The mathematical modeling of the anaerobic digestion process in landfills holds significant importance for multiple reasons. It's a fundamental tool for optimizing, and managing the generation of biogas, offering essential prospects for environmental sustainability and energy recovery.

In the literature, various models have been developed to describe the anaerobic digestion process and the associated microbial dynamics, aiming to better understand and optimize digester performance under varying microbial conditions^{5–13}. The Anaerobic Digestion Model No. 1 (ADM1) is a structural model that outlines the four main stages of anaerobic digestion leading to biogas production⁵. However, its complexity, due to a large number of equations, makes difficult its analysis. To overcome this, simplified models have been proposed. Rouez's model⁶ uses a two-step process (hydrolysis/acidogenesis and methanogenesis), and was mathematically analysed later by Ouchtout et al.¹⁴ highlighting the impact of the selected bacterial growth functions. In the context of wastewater treatment, the (AM2) model by Bernard et al.⁷ is notable, incorporating Monod growth for acidogenesis and Haldane growth for methanogenesis. This choice is well-suited for control and optimization, as demonstrated by the stability analysis conducted by Benyahia et al.¹⁵ Furthermore, in the context of co-digestion, where multiple types of substrates are mixed, Berga et al.¹⁶ have provided a mathematical analysis of Rakmak et al.'s two-step model (hydrolysis/acidogenesis and methanogenesis)⁸, and highlighting the influence of the chosen bacterial growth functions. However, the Halvadakis model⁹, which captures the natural process of anaerobic digestion, is of particular interest for our study. This model provides a comprehensive description of the three main stages: hydrolysis, acidogenesis, and methanogenesis. The model considers biodegradable organic matter (easily, intermediate, and slowly), which is hydrolyzed into a dissolved organic substrate. This substrate is then transformed into acetate substrate and carbon dioxide by the acidogenic biomass. The methanogenic biomass uses the acetate substrate to produce methane and additional carbon dioxide. The substrate utilization by biomass is determined by the Monod bacterial growth function. The model also takes into account the decomposition of dead acidogenic bacteria, which is converted to acetate and carbon dioxide, and the decomposition of dead methanogenic bacteria, which is converted to methane and carbon dioxide.

In this study, we perform a mathematical analysis of the Halvadakis dynamical system, incorporating bacterial growth functions $\mu_a(\cdot)$ and $\mu_m(\cdot)$, which can be of either the Monod or Haldane type^{17,18}. Regardless of the growth function selected, our analysis confirms the existence of an infinite number of non-hyperbolic equilibria toward which the system's trajectories converge. By employing Barbalat's lemma and the stable manifold theorem, we identify the attractor of the dynamical system and derive an expression for the biogas produced based on initial substrate concentrations and their limiting values. Through numerical simulation, we demonstrate how the process performance is influenced by the connectivity of the attractor when the Monod growth function is chosen, and by its non-connectivity when the Haldane growth function is applied. Furthermore, we demonstrate that biogas production, as a function of the initial stock concentration, exhibits five changes in monotonicity. These changes are driven by the four regions of convergence of substrate concentrations that result from the selection of the Haldane growth function. We also analyze the impact of bacterial mortality rates on biogas production, highlighting their significant role in the overall process.

Description of the model

In this section, we introduce the mathematical model for anaerobic digestion, inspired by the Halvadakis model. Initially, the biodegradable organic substrate present in the waste, with easily, intermediate, and slowly degradable components, breakdown into dissolved organic substrate with an hydrolysis constant k_h . The dissolved organic substrate is then consumed by acidogenic biomass at a rate of $\frac{1}{y_a}\mu_a(S_{dc})$. A portion of this consumption supports the growth of the acidogenic biomass, which is modeled as a natality rate $\mu_a(S_{dc})$ minus a mortality rate d_a . The remainder of the consumed substrate, expressed as $\frac{1 - y_a}{y_a}\mu_a(S_{dc})$, is partially converted to acetate substrate with a yield y_{ac} , while the rest is transformed into carbon dioxide with a yield of $(1 - y_{ac})$. A similar process applies to the acetate substrate and methanogenic biomass. The acetate substrate is consumed by methanogenic biomass at a rate of $\frac{1}{y_m}\mu_m(S_{ac})$. Part of this consumption is used for the growth of methanogenic biomass, described by a natality rate $\mu_m(S_{ac})$ minus a mortality rate d_m . The remaining portion, expressed as $\frac{1 - y_m}{y_m}\mu_m(S_{ac})$, leads to the production of methane with a yield y_{CH_4} , while the remainder produces carbon dioxide with a yield of $(1 - y_{CH_4})$. The model also emphasizes the significant role of mortality rates in biogas production. These rates are included in the equations for biogas production because they account for the natural decrease in biomass. This decrease influences substrate availability, which in turn affects the rates of methane

and carbon dioxide production. The mathematical model of the process described in Fig. 1 is formulated through a system of differential equations. This system is represented as follows:

$$\left\{ \begin{array}{l} \frac{dS_s}{dt} = -k_h S_s \\ \frac{dS_{dc}}{dt} = k_h S_s - \frac{1}{y_a} \mu_a(S_{dc}) X_a \\ \frac{dX_a}{dt} = (\mu_a(S_{dc}) - d_a) X_a \\ \frac{dS_{ac}}{dt} = y_{ac} \left(\frac{1 - y_a}{y_a} \mu_a(S_{dc}) + d_a \right) X_a - \frac{1}{y_m} \mu_m(S_{ac}) X_m \\ \frac{dX_m}{dt} = (\mu_m(S_{ac}) - d_m) X_m. \end{array} \right. \quad (1)$$

$$\left\{ \begin{array}{l} \frac{dCH_4}{dt} = y_{CH_4} \left(\frac{1 - y_m}{y_m} \mu_m(S_{ac}) + d_m \right) X_m \\ \frac{dCO_2}{dt} = (1 - y_{ac}) \left(\frac{1 - y_a}{y_a} \mu_a(S_{dc}) + d_a \right) X_a + (1 - y_{CH_4}) \left(\frac{1 - y_m}{y_m} \mu_m(S_{ac}) + d_m \right) X_m. \end{array} \right. \quad (2)$$

The bacterial growth functions denoted as μ_a and μ_m , are two C^1 -functions defined on $\mathbb{R}$ by:

- Monod law:

$$\mu_a(S_{dc}) = \frac{\mu_a^{max} S_{dc}}{K_d + S_{dc}} \text{ and } \mu_m(S_{ac}) = \frac{\mu_m^{max} S_{ac}}{K_a + S_{ac}}, \quad \forall S_{dc}, S_{ac} \in \mathbb{R}^+, \text{ or :}$$

- Haldane law:

$$\mu_a(S_{dc}) = \frac{\mu_a^{max} S_{dc}}{K_d + S_{dc} + \frac{S_{dc}^2}{K_{id}}} \text{ and } \mu_m(S_{ac}) = \frac{\mu_m^{max} S_{ac}}{K_a + S_{ac} + \frac{S_{ac}^2}{K_{ia}}}, \quad \forall S_{dc}, S_{ac} \in \mathbb{R}^+.$$

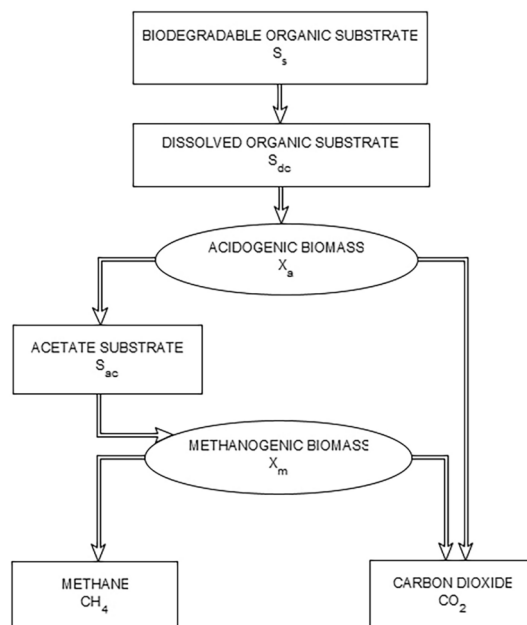


Fig. 1. Schematic representation of anaerobic digestion process in landfill.

- Haldane's law includes inhibition effects, providing a more realistic representation of bacterial growth at high substrate concentrations. Monod's growth function is appropriate for simpler systems with lower substrate concentrations and minimal inhibition effects, while the Haldane function is better suited for complex scenarios involving significant substrate inhibition, although it can affect biogas production. In the model, we have:

Variables:

S_s : Total concentration of biodegradable organic substrate (g/l),
 S_{dc} : Dissolved organic substrate concentration (g/l),
 X_a : Acidogenic biomass concentration (g/l),
 S_{ac} : Acetate substrate concentration (g/l),
 X_m : Methanogenic biomass concentration (g/l),
 CH_4 : Methane concentration (g/l),
 CO_2 : Carbon dioxide concentration (g/l).

Parameters:

y_a : mass conversion rate of dissolved organic substrate into acidogenic biomass,
 y_m : mass conversion rate of acetate substrate into methanogenic biomass,
 y_{ac} : acetate substrate conversion yield,
 y_{CH_4} : methane conversion yield.
 K_d : half-saturation constant of dissolved organic substrate (g/l),
 K_a : half-saturation constant of acetate substrate (g/l),
 K_{id} : inhibition constant of dissolved organic substrate (g/l),
 K_{ia} : inhibition constant of acetate substrate (g/l),
 k_h : hydrolysis constant of biodegradable organic substrate (day^{-1}),
 d_a : death rate of acidogenic bacteria (day^{-1}),
 d_m : death rate of methanogenic bacteria (day^{-1}).
 μ_a^{max} : maximum growth rate of acidogenic bacteria (day^{-1}),
 μ_m^{max} : maximum growth rate of methanogenic bacteria (day^{-1}).
 The model parameters satisfy the following conditions.

H1 . The substrate-biomass yields are strictly positives and:

$$y_a, y_m, y_{ac}, y_{CH_4} \in (0, 1).$$

H2 . The hydrolysis, half saturation and inhibition constants are strictly positives:

$$k_h, K_d, K_a, K_{id}, K_{ia} > 0.$$

H3 . The bacterial death rates are strictly positives and less than the maximum growth rates:

$$0 < d_a < \mu_a^{max} \text{ and } 0 < d_m < \mu_m^{max}.$$

We define the non-empty sets:

$$\xi_{dc} = \{s \in \mathbb{R}_+; \mu_a(s) \leq d_a\} \text{ and } \xi_{ac} = \{s \in \mathbb{R}_+; \mu_m(s) \leq d_m\}.$$

For the Monod expression, we have:

$$\xi_{dc} = [0, \lambda_a] \text{ with } \lambda_a = \frac{d_a K_d}{\mu_a^{max} - d_a}, \quad (3)$$

and

$$\xi_{ac} = [0, \lambda_m] \text{ with } \lambda_m = \frac{d_m K_a}{\mu_m^{max} - d_m}. \quad (4)$$

For the Haldane expression, we have:

$$\xi_{dc} = [0, \lambda_a^-] \cup [\lambda_a^+, +\infty) \text{ with } \lambda_a^\pm = \frac{\mu_a^{max} - d_a \pm \sqrt{\Delta_a}}{2d_a/K_{id}}, \quad (5)$$

and

$$\xi_{ac} = [0, \lambda_m^-] \cup [\lambda_m^+, +\infty) \text{ with } \lambda_m^\pm = \frac{\mu_m^{max} - d_m \pm \sqrt{\Delta_m}}{2d_m/K_{ia}}, \quad (6)$$

where

$$\Delta_a = (d_a - \mu_a^{max})^2 - 4 \frac{K_d}{K_{id}} d_a^2 \text{ and } \Delta_m = (d_m - \mu_m^{max})^2 - 4 \frac{K_a}{K_{ia}} d_m^2.$$

Study of the asymptotic behavior

As our system (1)–(2) satisfies the conditions of Cauchy-Lipschitz theorem, the existence and uniqueness of the solution with a given initial value is guaranteed. We begin with a result on the non-negativity and boundedness of the solution.

Proposition 1 Underhypothesis(H1-H3),foranynon-negativeinitialcondition $(S_s^0, S_{dc}^0, X_a^0, S_{ac}^0, X_m^0, CH_4^0, CO_2^0)$, the solution

$(S_s(\cdot), S_{dc}(\cdot), X_a(\cdot), S_{ac}(\cdot), X_m(\cdot), CH_4(\cdot), CO_2(\cdot))$ of (1)–(2) is non-negative and bounded.

Proof We will demonstrate the non-negativity of each component of the solution.

- For the component S_s , we have $\frac{dS_s}{dt} = -k_h S_s$ with $S_s(0) = S_s^0$, which implies $S_s(t) = S_s^0 e^{-k_h t}$. If $S_s^0 \geq 0$, then for all $t \geq 0$, we have $S_s(t) \geq 0$.
- For the component S_{dc} , we have $\frac{dS_{dc}}{dt} = k_h S_s - \frac{1}{y_a} \mu_a(S_{dc}) X_a$ with $S_{dc}(0) = S_{dc}^0$. We have $\frac{dS_{dc}}{dt} \geq -\frac{1}{y_a} \mu_a(S_{dc}) X_a$ for all $t \geq 0$. According to the comparison theorem¹⁹, if $S_{dc}^0 > \underline{S}$, then $S_{dc}(t) > \underline{S}(t)$, such that $\underline{S}(t)$ is a solution of the differential equation:

$$\frac{d\underline{S}}{dt} = -\frac{1}{y_a} \mu_a(\underline{S}) X_a, \quad \underline{S}(0) = S_{dc}^0. \quad (7)$$

If $S_{dc}^0 = 0$, then $\underline{S} \equiv 0$ is the only solution of (7), thus $\underline{S}(t) > 0$ when $S_{dc}^0 > 0$ for all $t \geq 0$. Then, if $S_{dc}^0 > 0$, we have $S_{dc}(t) > 0$ for all $t \geq 0$.

- For the component X_a , we have $\frac{dX_a}{dt} = (\mu_a(S_{dc}) - d_a) X_a$ with $X_a(0) = X_a^0$. The term $(\mu_a(S_{dc}) - d_a) X_a$ is linear with respect to X_a , thus if $X_a^0 \geq 0$, we have $X_a(t) \geq 0$ for all $t \geq 0$.
- For the component S_{ac} , we have $\frac{dS_{ac}}{dt} = y_{ac} \left(\frac{1 - y_a}{y_a} \mu_a(S_{dc}) + d_a \right) X_a - \frac{1}{y_m} \mu_m(S_{ac}) X_m$ with $S_{ac}(0) = S_{ac}^0$. We have $\frac{dS_{ac}}{dt} \geq -\frac{1}{y_m} \mu_m(S_{ac}) X_m$ for all $t \geq 0$. According to the comparison theorem, if $S_{ac}^0 > \bar{S}$, then $S_{ac}(t) > \bar{S}(t)$, such that $\bar{S}(t)$ is a solution of the differential equation:

$$\frac{d\bar{S}}{dt} = -\frac{1}{y_m} \mu_m(\bar{S}) X_m, \quad \bar{S}(0) = S_{ac}^0. \quad (8)$$

If $S_{ac}^0 = 0$, then $\bar{S} \equiv 0$ is the only solution of (8), thus $\bar{S}(t) > 0$ when $S_{ac}^0 > 0$ for all $t \geq 0$. Then, if $S_{ac}^0 > 0$, we have $S_{ac}(t) > 0$ for all $t \geq 0$.

- For the component X_m , we have $\frac{dX_m}{dt} = (\mu_m(S_{ac}) - d_m) X_m$ with $X_m(0) = X_m^0$. The term $(\mu_m(S_{ac}) - d_m) X_m$ is linear with respect to X_m , thus if $X_m^0 \geq 0$, we have $X_m(t) \geq 0$ for all $t \geq 0$.
- From the above, we deduce the non-negativity of the components CH_4 and CO_2 . For boundedness, it suffices to notice that our system is closed, that is:

$$\frac{dS_s}{dt} + \frac{dS_{dc}}{dt} + \frac{dX_a}{dt} + \frac{dS_{ac}}{dt} + \frac{dX_m}{dt} + \frac{dCH_4}{dt} + \frac{dCO_2}{dt} = 0,$$

so, there exists a positive constant C such that for all $t \geq 0$, we have:

$$S_s(t) + S_{dc}(t) + X_a(t) + S_{ac}(t) + X_m(t) + CH_4(t) + CO_2(t) = C. \quad (9)$$

Hence, the result is proven. $\square$

Next, we proceed to investigate the asymptotic behaviour of (1).

Proposition 2 Under hypothesis (H1–H3), for any non-negative initial condition $(S_s^0, S_{dc}^0, X_a^0, S_{ac}^0, X_m^0)$, the solution of (1) verifies:

1. $\lim_{t \rightarrow +\infty} S_s(t) = \lim_{t \rightarrow +\infty} X_a(t) = \lim_{t \rightarrow +\infty} X_m(t) = 0$.
2. There exist two positive real numbers S_{dc}^* and S_{ac}^* , which satisfy:

$$\lim_{t \rightarrow +\infty} S_{dc}(t) = S_{dc}^* \text{ and } \lim_{t \rightarrow +\infty} S_{ac}(t) = S_{ac}^*.$$

Proof We have $\lim_{t \rightarrow +\infty} S_s(t) = \lim_{t \rightarrow +\infty} S_s^0 e^{-k_h t} = 0$. To prove the convergence of X_a and S_{dc} , we consider the C^1 -function defined on $[0, +\infty)$ as follows:

$$V_1(t) = S_s(t) + \frac{1}{y_a} X_a(t) + S_{dc}(t).$$

We have:

$$\frac{dV_1}{dt} = -\frac{d_a}{y_a} X_a.$$

Since X_a is non-negative, then V_1 is non-increasing, and minored by 0. So there exists a real $l_1 \geq 0$ such that $\lim_{t \rightarrow +\infty} V_1(t) = l_1$ ($l_1 < \infty$). Moreover, one has:

$$\frac{d^2 V_1}{dt^2} = -\frac{d_a}{y_a} (\mu_a(S_{dc}) - d_a) X_a.$$

Since $\frac{d^2 V_1}{dt^2}$ is bounded, $\frac{dV_1}{dt}$ is uniformly continuous. According to Barbalat's lemma²⁰, we have $\lim_{t \rightarrow +\infty} \frac{dV_1}{dt}(t) = 0$. Then $\lim_{t \rightarrow +\infty} X_a(t) = 0$.

Additionally, since $\lim_{t \rightarrow +\infty} V_1(t) = \lim_{t \rightarrow +\infty} S_s(t) + \frac{1}{y_a} X_a(t) + S_{dc}(t) = l_1$, we have $\lim_{t \rightarrow +\infty} S_{dc}(t) = S_{dc}^* = l_1$. Similarly for X_m and S_{ac} , we consider the C^1 -function defined on $[0, +\infty)$ as follows:

$$V_2(t) = S_{ac}(t) + \frac{1}{y_m} X_m(t) + y_{ac} (S_s(t) + X_a(t) + S_{dc}(t)),$$

we prove that $\lim_{t \rightarrow +\infty} X_m(t) = 0$ and $\lim_{t \rightarrow +\infty} S_{ac}(t) = S_{ac}^*$. $\square$

Proposition 3 Under hypothesis (H1–H3), for any non-negative initial condition $(S_s^0, S_{dc}^0, X_a^0, S_{ac}^0, X_m^0)$, the solution of (1) converges asymptotically to an equilibrium $(0, S_{dc}^*, 0, S_{ac}^*, 0)$, where S_{dc}^* and S_{ac}^* belong respectively to ξ_{dc} and ξ_{ac} .

Proof Let $S_{dc}(\cdot)$ be a component of the solution of (1) which converges to S_{dc}^* , and suppose that S_{dc}^* doesn't belong to ξ_{dc} .

Since μ_a is continuous on $[0, +\infty)$, then $\mu_a(S_{dc}(t))$ tends to $\mu_a(S_{dc}^*)$ when t approaches $+\infty$, leading to:

$$\forall \epsilon > 0, \quad \exists T \geq 0 : \quad \forall t > T, \quad -\epsilon < \mu_a(S_{dc}(t)) - \mu_a(S_{dc}^*) < \epsilon.$$

Choosing $\epsilon = \frac{\mu_a(S_{dc}^*) - d_a}{2} > 0$, we have, for all $t > T$:

$$\begin{aligned} \mu_a(S_{dc}(t)) - \mu_a(S_{dc}^*) &> -\epsilon \Rightarrow \mu_a(S_{dc}(t)) - \mu_a(S_{dc}^*) + 2\epsilon > \epsilon \\ &\Rightarrow \mu_a(S_{dc}(t)) - d_a > \epsilon \\ &\Rightarrow \frac{dX_a}{dt}(t) > \epsilon X_a(t). \end{aligned}$$

Letting $s \in (T, t)$, integrating from s to t , we get $X_a(t) > X_a(s)e^{\epsilon(t-s)}$, for all $t > T$. This contradicts the fact that X_a is bounded. Thus, we conclude that S_{dc}^* belongs to ξ_{dc} .

Similarly, we prove that S_{ac}^* belongs to ξ_{ac} . $\square$

The sub-system (1) has an infinite number of equilibria that are non-hyperbolic, therefore the linearization does not allow us to conclude stability. To overcome this difficulty, we carry out a change of variables that leads to the use of a hyperbolic system. We then apply the stable manifold theorem. However, we obtain the following result.

Proposition 4 Under hypothesis (H1–H3), for each steady state $E = (0, S_{dc}^*, 0, S_{ac}^*, 0)$ with $S_{dc}^* \in \xi_{dc}$ and $S_{ac}^* \in \xi_{ac}$, there exists an invariant three-dimensional manifold $\mathbb{M}$ in $\mathbb{R}_+^5$ such that any solution of (1) with an initial condition in $\mathbb{M}$ converges asymptotically to E .

Proof Let $(S_s, S_{dc}, X_a, S_{ac}, X_m)$ be a solution of (1) with an initial condition in $\mathbb{M}$. Let Z_1 and Z_2 be the two variables defined by:

$$Z_1(t) = \frac{S_s(t) + S_{dc}(t) - S_{dc}^*}{X_a(t)} + \beta_1, \quad \forall t \geq 0,$$

and

$$Z_2(t) = \frac{y_{ac}(S_s(t) + S_{dc}(t) + X_a(t)) + S_{ac}(t) - y_{ac}S_{dc}^* - S_{ac}^*}{X_m(t)} + \beta_2, \quad \forall t \geq 0,$$

with

$$\beta_1 = \frac{\mu_a(S_{dc}^*)}{y_a(\mu_a(S_{dc}^*) - d_a)} \text{ and } \beta_2 = \frac{\mu_m(S_{ac}^*)}{y_m(\mu_m(S_{ac}^*) - d_m)}.$$

We have:

$$\begin{aligned} \frac{dZ_1}{dt} &= \frac{-\frac{1}{y_a}\mu_a(S_{dc})X_a^2 - (Z_1 - \beta_1)(\mu_a(S_{dc}) - d_a)X_a^2}{X_a^2} \\ &= \mu_a(S_{dc})(\beta_1 - \frac{1}{y_a}) - \beta_1 d_a - (\mu_a(S_{dc}) - d_a)Z_1 \\ &= -\gamma_1(\mu_a(S_{dc}) - \mu_a(S_{dc}^*)) - (\mu_a(S_{dc}) - d_a)Z_1, \text{ with } \gamma_1 = -\frac{d_a}{y_a(\mu_a(S_{dc}^*) - d_a)}, \end{aligned}$$

and

$$\begin{aligned} \frac{dZ_2}{dt} &= \frac{-\frac{1}{y_m}\mu_m(S_{ac})X_m^2 - (Z_2 - \beta_2)(\mu_m(S_{ac}) - d_m)X_m^2}{X_m^2} \\ &= \mu_m(S_{ac})(\beta_2 - \frac{1}{y_m}) - \beta_2 d_m - (\mu_m(S_{ac}) - d_m)Z_2 \\ &= -\gamma_2(\mu_m(S_{ac}) - \mu_m(S_{ac}^*)) - (\mu_m(S_{ac}) - d_m)Z_2, \text{ with } \gamma_2 = -\frac{d_m}{y_m(\mu_m(S_{ac}^*) - d_m)}. \end{aligned}$$

If we define $Y = (S_s, Z_1, X_a, Z_2, X_m)$, then we have:

$$\frac{dZ_1}{dt} = -\gamma_1(g_a(Y) - \mu_a(S_{dc}^*)) - (g_a(Y) - d_a)Z_1,$$

and

$$\frac{dZ_2}{dt} = -\gamma_2(g_m(Y) - \mu_m(S_{ac}^*)) - (g_m(Y) - d_m)Z_2,$$

with

$$\begin{aligned} g_a(Y) &= \mu_a(S_{dc}) = \mu_a((Z_1 - \beta_1)X_a - S_s + S_{dc}^*) \text{ and} \\ g_m(Y) &= \mu_m(S_{ac}) = \mu_m((Z_2 - \beta_2)X_m - y_{ac}(Z_1 - \beta_1 + 1)X_a + S_{ac}^*). \end{aligned}$$

The equivalent system of the system (1) in the domain $\mathbb{D} = \{(S_s, Z_1, X_a, Z_2, X_m) \in \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}_+^* \times \mathbb{R} \times \mathbb{R}_+^* / (Z_1 - \beta_1)X_a - S_s + S_{dc}^* \geq 0; (Z_2 - \beta_2)X_m - y_{ac}(Z_1 - \beta_1 + 1)X_a \geq 0\}$ is written:

$$\left\{ \begin{array}{l} \frac{dS_s}{dt} = -k_h S_s \\ \frac{dX_a}{dt} = (g_a(Y) - d_a) X_a \\ \frac{dX_m}{dt} = (g_m(Y) - d_m) X_m \\ \frac{dZ_1}{dt} = -\gamma_1(g_a(Y) - \mu_a(S_{dc}^*)) - (g_a(Y) - d_a) Z_1 \\ \frac{dZ_2}{dt} = -\gamma_2(g_m(Y) - \mu_m(S_{ac}^*)) - (g_m(Y) - d_m) Z_2. \end{array} \right. \quad (10)$$

The domain $\mathbb{D}$ is positively invariant for the system (10). Since $\mu_a(S_{dc}^*) \neq d_a$ and $\mu_m(S_{ac}^*) \neq d_m$, 0 is the only equilibrium of (10) on $\mathbb{D}$. The Jacobian matrix is given by:

$$J(0) = \begin{bmatrix} -k_h & 0 & 0 & 0 & 0 \\ 0 & \mu_a(S_{dc}^*) - d_a & 0 & 0 & 0 \\ 0 & 0 & \mu_m(S_{ac}^*) - d_m & 0 & 0 \\ \gamma_1 \mu_a'(S_{dc}^*) & \gamma_1 \beta_1 \mu_a'(S_{dc}^*) & 0 & -(\mu_a(S_{dc}^*) - d_a) & 0 \\ 0 & \gamma_2 y_{ac}(1 - \beta_2) \mu_m'(S_{ac}^*) & \gamma_2 \beta_2 \mu_m'(S_{ac}^*) & 0 & -(\mu_m(S_{ac}^*) - d_m) \end{bmatrix}$$

We have three negative eigenvalues: $-k_h$, $\mu_a(S_{dc}^*) - d_a$ and $\mu_m(S_{ac}^*) - d_m$, and two positive eigenvalues: $-(\mu_a(S_{dc}^*) - d_a)$ and $-(\mu_m(S_{ac}^*) - d_m)$. By the stable manifold theorem²¹, 0 is a hyperbolic equilibrium with a three-dimensional stable manifold $\mathbb{W}^s$ and a two-dimensional unstable manifold $\mathbb{W}^u$.

We conclude that any trajectory of (10) with an initial value in the three-dimensional invariant manifold $\mathbb{M} = \mathbb{W}^s \cap \mathbb{D}$ converges asymptotically to 0, therefore the corresponding trajectory of (1) converges asymptotically to $E = (0, S_{dc}^*, 0, S_{ac}^*, 0)$. $\square$

We can now state the corollary.

Corollary 1 Under hypothesis (H1–H3), the set:

$$\{0\} \times \xi_{dc} \times \{0\} \times \xi_{ac} \times \{0\}$$

is an attractor of sub-system (1).

Let's proceed to the sub-system (2).

Proposition 5 Under hypothesis (H1–H3), for any non-negative initial condition $(S_s^0, S_{dc}^0, X_a^0, S_{ac}^0, X_m^0, CH_4^0, CO_2^0)$, the solution of (2) verifies:

$$\lim_{t \rightarrow +\infty} CH_4(t) = CH_4^0 + y_{CH_4}(\alpha - S_{ac}^* - y_{ac} S_{dc}^*),$$

and

$$\lim_{t \rightarrow +\infty} CO_2(t) = CO_2^0 + \beta - \alpha y_{CH_4} + (y_{ac} y_{CH_4} - 1) S_{dc}^* + (y_{CH_4} - 1) S_{ac}^*,$$

with

$$\alpha = y_{ac}(S_s^0 + S_{dc}^0 + X_a^0) + S_{ac}^0 + X_m^0 \text{ and } \beta = S_s^0 + S_{dc}^0 + X_a^0 + S_{ac}^0 + X_m^0.$$

Proof From (2), we have:

$$\begin{aligned} \frac{dCO_2}{dt} &= (1 - y_{CH_4}) \left(\frac{1 - y_m}{y_m} \mu_m(S_{ac}) + d_m \right) X_m + (1 - y_{ac}) \left(\frac{1 - y_a}{y_a} \mu_a(S_{dc}) + d_a \right) X_a \\ &= \frac{1 - y_{CH_4}}{y_{CH_4}} \times \frac{dCH_4}{dt} + (1 - y_{ac}) \left(\frac{1 - y_a}{y_a} \mu_a(S_{dc}) + d_a \right) X_a \\ &= \frac{1 - y_{CH_4}}{y_{CH_4}} \times \frac{dCH_4}{dt} - (1 - y_{ac}) \left(\frac{dS_s}{dt} + \frac{dS_{dc}}{dt} + \frac{dX_a}{dt} \right). \end{aligned}$$

For all $t \geq 0$, by integrating from 0 to t , we get:

$$CO_2(t) - CO_2^0 = \frac{1 - y_{CH_4}}{y_{CH_4}} (CH_4(t) - CH_4^0) - (1 - y_{ac})(S_s(t) - S_s^0 + S_{dc}(t) - S_{dc}^0 + X_a(t) - X_a^0).$$

Let's denote $\lim_{t \rightarrow +\infty} CH_4(t) = \widehat{CH_4}$ and $\lim_{t \rightarrow +\infty} CO_2(t) = \widehat{CO_2}$. Upon taking limits, we obtain:

$$\widehat{CO_2} - \frac{1 - y_{CH_4}}{y_{CH_4}} \widehat{CH_4} = CO_2^0 - \frac{1 - y_{CH_4}}{y_{CH_4}} CH_4^0 - (1 - y_{ac})(S_{dc}^* - S_s^0 - S_{dc}^0 - X_a^0). \quad (11)$$

On the other hand, from Eq. (9), for all $t \geq 0$, we have:

$$S_s(t) + S_{dc}(t) + X_a(t) + S_{ac}(t) + X_m(t) + CH_4(t) + CO_2(t) = S_s^0 + S_{dc}^0 + X_a^0 + S_{ac}^0 + X_m^0 + CH_4^0 + CO_2^0,$$

when t tends to $+\infty$, we obtain:

$$S_{dc}^* + S_{ac}^* + \widehat{CH_4} + \widehat{CO_2} = S_s^0 + S_{dc}^0 + X_a^0 + S_{ac}^0 + X_m^0 + CH_4^0 + CO_2^0. \quad (12)$$

From Eq. (11) and Eq. (12), we conclude that:

$$\begin{aligned} \widehat{CH_4} &= y_{CH_4} (X_m^0 + S_{ac}^0 + y_{ac}(S_s^0 + S_{dc}^0 + X_a^0) - S_{ac}^* - y_{ac}S_{dc}^*) + CH_4^0, \text{ and} \\ \widehat{CO_2} &= -\widehat{CH_4} - S_{dc}^* - S_{ac}^* + S_s^0 + S_{dc}^0 + X_a^0 + S_{ac}^0 + X_m^0 + CH_4^0 + CO_2^0. \\ \text{Taking } \alpha &= y_{ac}(S_s^0 + S_{dc}^0 + X_a^0) + S_{ac}^0 + X_m^0 \text{ and } \beta = S_s^0 + S_{dc}^0 + X_a^0 + S_{ac}^0 + X_m^0, \text{ we obtain the result. } \square \end{aligned}$$

Numerical results

In this section, we utilize the Matlab programming language to display numerical results for quantities of biogas produced, impacted by the final values S_{dc}^* and S_{ac}^* of the concentrations of the dissolved organic and acetate substrates.

The parameter values used in our numerical simulations are inspired by^{10,11,14} and are presented in the following tables:

We assume that at the beginning of the process, there is no production of biogas, so that $CH_4^0 = CO_2^0 = 0$, the other initial conditions are given in Table 1.

Figures 2 and 3 describe the trajectories of the system (1)–(2) corresponding to the Monod and Haldane scenarios, respectively.

In the Monod case, the concentrations of carbon dioxide and methane reach stability at approximately 200g/l and 250g/l, respectively, following a storage period of 200 days. Notably, an observation can be made that the anaerobic phase begins at around 100 days, signifying a switch wherein methane production exceeds carbon dioxide production.

In the Haldane case, the stabilization of carbon dioxide and methane production happens nearly simultaneously but after an extended storage period of 400 days. The start of the anaerobic phase is observed after approximately 300 days in this scenario.

The disparity between the Monod and Haldane cases lies in the convergence behavior of the concentrations S_{dc} and S_{ac} . In the Monod case, the concentrations converge within connected sets, corresponding to the connected attractor. In contrast, the Haldane case shows convergence within non-connected sets, corresponding to the non-connected attractor. To validate this assertion, we fix the parameters specified in Table 2 while varying the initial values. We, then, observe the convergence of the concentrations S_{dc} and S_{ac} .

In the Monod case, from Eq. (3) and Eq. (4), we have $\lambda_a = 5.5$ and $\lambda_m = 60$.

Figures 4 and 5 illustrate that for any non-negative value of S_s^0 , S_{dc} and S_{ac} converge to values less than λ_a and λ_m respectively.

In the Haldane case, Eq. (5) yields $\lambda_a^- = 5.59$ and $\lambda_a^+ = 894.4$. Figure 6 demonstrates that when:

- $S_s^0 \leq 3300$, S_{dc} converges to values less than λ_a^- .
- $S_s^0 > 3300$, S_{dc} converges to values higher than λ_a^+ . Therefore, we find that 3300 is the critical value of S_s^0 at which the limit S_{dc}^* shifts from the interval λ_a^- to the interval λ_a^+ .

Similarly, Eq. (6) gives $\lambda_m^- = 123.24$ and $\lambda_m^+ = 243.42$. Figure 7 shows that when:

- $0 \leq S_s^0 \leq 600$, S_{ac} converges to values less than λ_m^- .
- $600 < S_s^0 < 3900$, S_{ac} converges to values higher than λ_m^+ .
- $S_s^0 \geq 3900$, S_{ac} converges again to values less than λ_m^- . Therefore, we find that 600 and 3900 are critical values of S_s^0 at which the limit S_{ac}^* shifts from λ_m^- to λ_m^+ and then back to λ_m^- .

S_s^0	S_{dc}^0	X_a^0	S_{ac}^0	X_m^0
400	50	15	10	1, 5

Table 1. Initial conditions.

μ_a^{max}	K_d	y_a	d_a	μ_m^{max}	K_a	y_m	d_m	k_h	y_{ac}	y_{CH_4}	K_{id}	K_{ih}
0, 5	50	0, 15	0, 05	0, 25	600	0, 06	0, 03	0, 02	0, 7	0, 8	100	50

Table 2. Model coefficients.

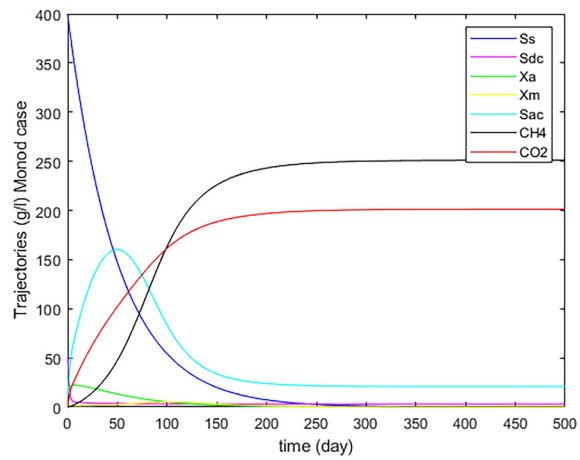


Fig. 2. Trajectories of the system using the Monod growth function.

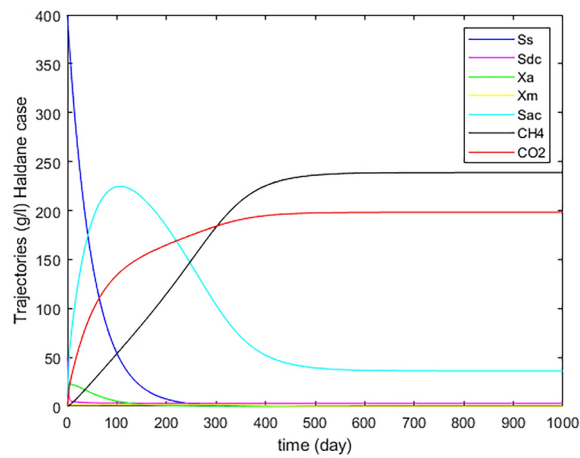


Fig. 3. Trajectories of the system using the Haldane growth function.

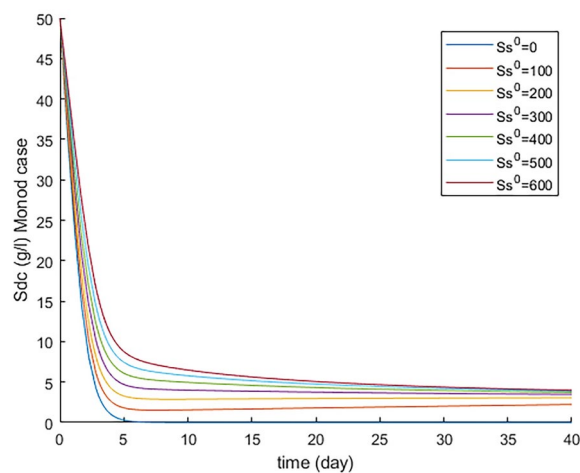


Fig. 4. S_{dc} trajectories for various values of S_s^0 in the Monod case.

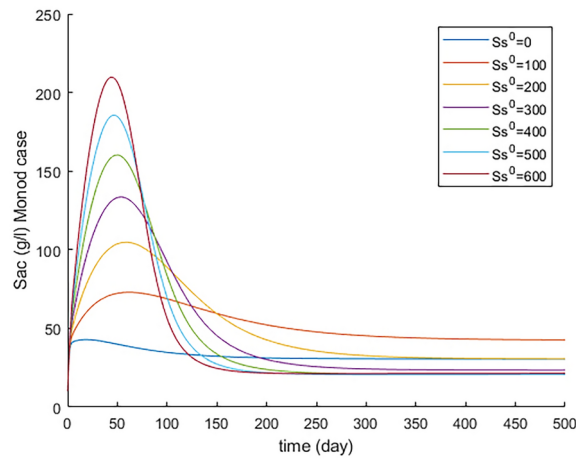


Fig. 5. S_{ac} trajectories for various values of S_s^0 in the Monod case.

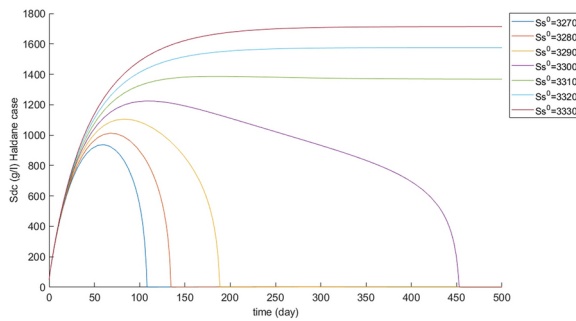


Fig. 6. S_{dc} trajectories for various values of S_s^0 in the Haldane case.

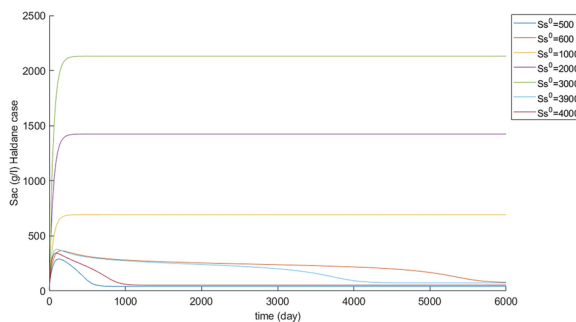


Fig. 7. S_{ac} trajectories for various values of S_s^0 in the Haldane case.

When the Haldane growth function is considered, we observe two regions of convergence of S_{dc} and two regions of convergence of S_{ac} . In Fig. 8, we illustrate the behavior of the substrate concentrations within the dynamic model (1) and identify the placement of these four regions. Considering initial substrate concentrations S_s^0 ranging from 0 to 6000 with an increment of 200, this range includes the switch points 600 and 3900 for S_{ac}^* and 3300 for S_{dc}^* . We observe the following:

- When $S_s^0 \leq 600$, both S_{dc}^* and S_{ac}^* are below λ_a^- and λ_m^- respectively.
- For $600 < S_s^0 \leq 3300$, S_{dc}^* remains below λ_a^- , while S_{ac}^* surpasses λ_m^+ .
- Between $3300 < S_s^0 < 3900$, both S_{dc}^* and S_{ac}^* exceed λ_a^+ and λ_m^+ respectively.
- When $S_s^0 \geq 3900$, S_{dc}^* rises above λ_a^+ , while S_{ac}^* falls below λ_m^- .

This figure reveals the impact of inhibition phenomena on the process. Specifically, inhibition can prevent the bacteria from efficiently consuming the substrate, resulting in high concentrations substrate remaining in the

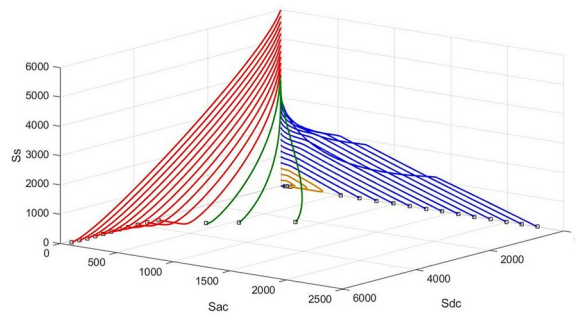


Fig. 8. S_s , S_{dc} , and S_{ac} trajectories behavior for various values of S_s^0 in the Haldane case.

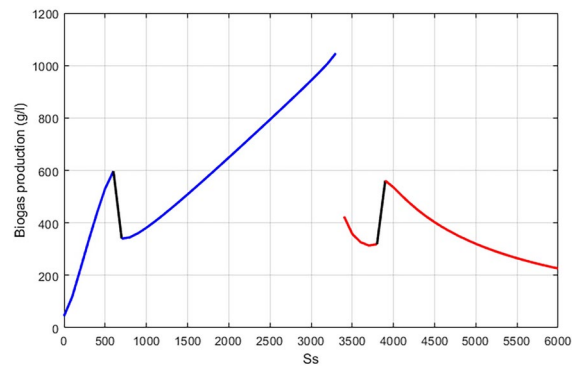


Fig. 9. Biogas production as function of initial stock concentration S_s^0 .

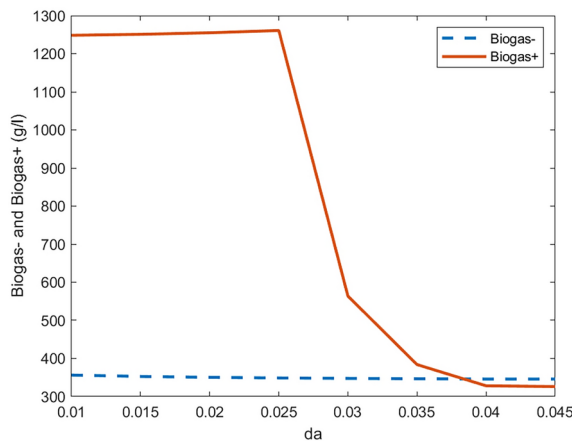


Fig. 10. Biogas production as function of mortality rate d_a .

process. As a result, the accumulation of unconsumed substrate affects negatively biogas production, as will be further illustrated in the next figure. In contrast, when the Monod growth function is chosen, only one region of convergence is observed, and the remaining substrate does not reach a high concentration.

In the following, we will focus on simulations in the Haldane case.

Figure 9 shows the total biogas (methane + carbon dioxide) produced as a function of the initial substrate concentration S_s^0 . According to Proposition 5, the quantity of biogas produced is influenced by the limits S_{dc}^* and S_{ac}^* , which shift from lower to higher values as S_s^0 varies. In this figure, we observe that for $0 \leq S_s^0 \leq 3300$, S_{dc} converges below λ_a^- . This indicates that the dissolved organic substrate is effectively consumed by acidogenic bacteria, resulting in a significant increase in biogas production, which reaches a peak concentration of 1050 g/l. However, when $S_s^0 > 3300$, S_{dc} converges above λ_a^+ , leading to less effective consumption of the dissolved organic substrate by acidogenic bacteria. Consequently, biogas production declines sharply, with a noticeable

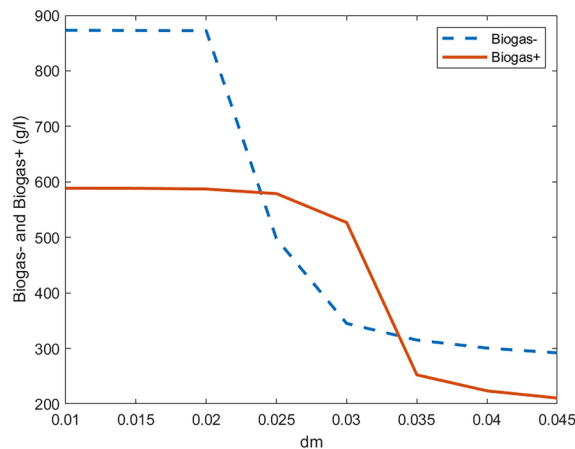


Fig. 11. Biogas production as function of mortality rate d_m .

fall between 3300 and 3400 in S_s^0 , where biogas output falls to 400 g/l. Beyond this point, biogas concentration continues to decrease. Within this range, for $S_s^0 > 600$, S_{dc} converges above λ_m^+ , resulting in poor consumption of acetate substrate by methanogenic bacteria, which leads to a decrease in biogas concentration over a small interval, followed by a gradual increase. Similarly, for $S_s^0 > 3900$, S_{dc} converges below λ_m^- , leading to effective consumption of acetate substrate by methanogenic bacteria and a temporary increase in biogas concentration, which is then followed by a continuous decrease.

These observations highlight that the choice of growth functions significantly affects biogas production. When both bacterial growth functions are Haldane, biogas production exhibits five distinct switches in monotonicity. In contrast, when one Monod function and one Haldane function are used, there is a single switch value of S_s^0 at which biogas production exhibits one change in monotonicity, shifting from increasing to decreasing. When both functions are Monod, biogas production increases linearly with S_s^0 without any decrease. This occurs because there is no inhibition phenomenon present, allowing the bacteria to continuously consume the substrate.

Let's examine the impact of bacterial mortality rates on biogas production by considering a small value of S_s^0 yielding Biogas- and a large value of S_s^0 yielding Biogas+. It can be seen in Fig. 10 that when the initial stock concentration S_s^0 is less than 3300, the mortality rate of acidogenic bacteria has almost no influence on the biogas production. In the other hand, Fig. 11 shows that the amount of biogas is first constant and then decreases as function of the mortality rate of methanogenic bacteria.

When S_s^0 is higher than 3400, both Figs. 10 and 11 illustrate that the biogas production remains constant when the mortality rates of acidogenic and methanogenic bacteria are below 0.025. However, it starts decreasing once the mortality rates surpass this value. Notably, in this scenario, biogas production decreases sharply with mortality rate of acidogenic bacteria.

Conclusion

Our study has provided a detailed analysis of biogas production through anaerobic digestion of solid waste, modeled by a system of differential equations. Through our mathematical analysis, we identified the attractor of the system. Our numerical simulations provide important insights into system behavior. When both bacterial growth functions are Monod, the attractor is connected, and biogas production increases linearly with the initial stock concentration S_s^0 . When one growth function is Monod and the other is Haldane, the attractor becomes non-connected with two regions of convergence, and biogas production as function of S_s^0 exhibits a single change in monotonicity. When both functions are Haldane, the attractor is non-connected with four regions of convergence, and biogas production as function of S_s^0 exhibits five changes in monotonicity. Additionally, biogas production decreases with higher bacterial mortality rates.

In our future work, we will focus on managing the initial stock concentration to avoid exceeding the switch value of S_s^0 that results in optimal biogas production and will explore the impact of spatial diffusion effects on biogas production.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

I.A. analyzed the outcomes. A.A. and I.A. conducted the numerical simulations. N.E.K. and A.A. identified the problem and provided supervision. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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